

Multimodal Exploration of Terrain with Immersive Interactions

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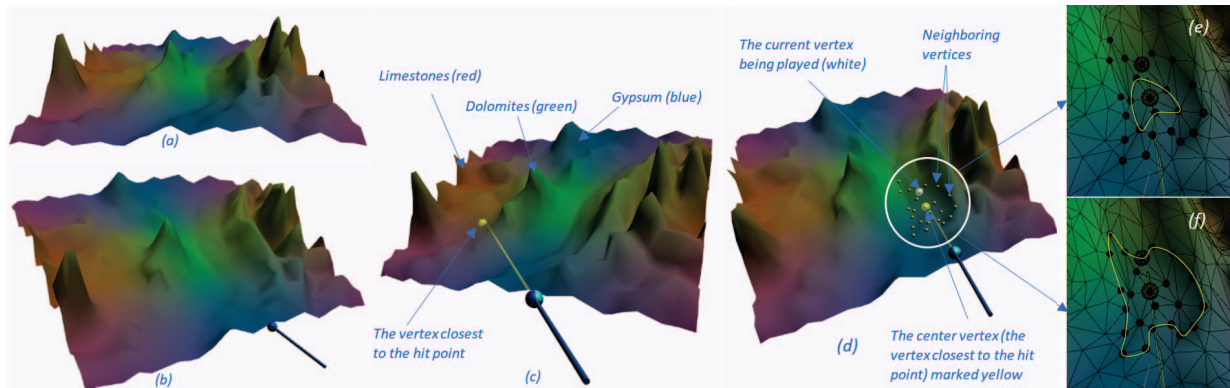


Figure 1: (a) A terrain model; (b) a terrain model being grabbed in a virtual environment (VE); (c) in the “Continuous” mode, a ray was cast on the terrain to generate audio feedback for the underlying rock type where the ray intersected the terrain; (d) in the “Sporadic” mode, as a ray was cast, a small region on the terrain resonated with different pitches of music (the vertex closest to the ray intersection point, called the center vertex, shown in yellow); (e) the neighboring vertices closest to the center vertex was connected with a yellow line; (f) the neighboring vertices further from the center vertex was connected with a yellow line.

ABSTRACT

Virtual reality (VR) lets users explore objects in an immersive environment. Users freely interact with objects in a virtual environment (VE), where they can assign physical properties to objects, make them animate or inanimate, and have them respond according to the underlying characteristics. This research explores geological topology using VR. In addition, users studied a terrain model in a multimodal audio-visual VE. Initial findings showed that users preferred multimodal exploration of terrain models compared to the traditional unimodal vision-only version. Two audio-enhanced modes were used: “Continuous” and “Sporadic.”

Index Terms: Human-centered computing [Visualization techniques]; Multimodality; Human-centered computing [Visualization]; Visualization design and evaluation methods; Human-centered computing [Visualization]; Virtual reality

1 INTRODUCTION

Multi-sensory systems have been proven to be superior to traditional visual feedback-based virtual systems in terms of the sense of presence and user preference [3]. Audio-visual cartography with an immersive virtual environment (VE) enriched user experience [2]. Sonification facilitated visual perception and helped users overcome challenges in visual representations [4]. Visualizer [1] was developed for interactive visual exploration of geoscience in immersive VEs. This research investigated multimodality through a virtual reality (VR)-enhanced geological topology exploration. A concise user study was carried out that demonstrated users preferred the audio-visual mode over the unimodal vision-only option.

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2 THE PROPOSED APPROACH

A karst terrain featuring a highly undulated topology (Figure 1) was used. Karsts mainly comprise three rocks: limestone, dolomite, and gypsum. Underlying rocks were colored differently to differentiate from each other. For ease of differentiation, limestones were colored red, dolomites looked green, and the gypsum concentration was blue. In addition to exploring the topological differences in the terrain model, users got a better understanding of underlying rocks from the assigned colors. Users explored a 3D terrain model of karst in VR by grabbing, moving, and positioning it with different orientations (Figure 1 b). Next, users performed an audio-visual exploration.

The audio-visual mode used two options: continuous versus sporadic. In the continuous mode, users cast rays on the model to explore the underlying rocks (Figure 1 c). Depending on the rock types, music was played at different pitches as rays were cast. For limestones, a high pitch was chosen, green-colored dolomites generated a medium pitch, whereas blue gypsum responded with a low pitch. In the sporadic mode, as a ray was cast, a small region surrounding the intersection point resonated with different pitches depending on the rock types. This is demonstrated in Figure 1 d.

3 IMPLEMENTATION

The project used Unity3d for its implementation. An Oculus Quest 2 headset was used for VR integration. With the unimodal vision-only option, the user grabbed the terrain model and moved it closer or further to understand variations in the underlying rocks. In the audio-visual mode, two buttons called, “Continuous” and “Sporadic,” toggled on and off when pressed. In the continuous mode, as a ray was cast by pressing the trigger button of the VR controller, the vertex closest to the intersection point (shown as a yellow sphere in Figure 1 c) generated audio feedback with a pitch value that varied with the underlying rock type, as explained in the previous section.

With the sporadic option, the vertex closest to the intersection point was called the center vertex. The neighboring vertices surrounding the center vertex enclosing a small region were identified. All these vertices were marked as yellow spheres, shown in Figure

1 d to Figure 1 f. The center vertex was enclosed in a larger transparent yellow sphere to make it distinct from neighboring vertices. Starting from the center vertex, depending on the underlying rocks, each vertex played music with the assigned pitch for 5 seconds. The current playing vertex was enclosed in a transparent white sphere, as shown in Figure 1d. The musical pitches in this region continued repeating until the user cast a ray onto another part of the scene or stopped sporadic exploration by pressing the corresponding button. The sporadic mode helped users understand the underlying variations in rocks within a region that could be made smaller or larger. The smallest zone included only the center vertex. A larger zone expanded outward to include neighbors of the center vertex, neighbors of the neighboring vertices, and so on.

4 EVALUATION

Recent research demonstrated participants stated performing tasks faster in the audiovisual mode compared to the vision-only option, though the measured response time was longer in the multimodal display compared to the vision-only version [4]. Wang et al. [5] ignored time but incorporated emotion to characterize the value of visualization. Considering the overall aspect of multisensory visualization, subjective measures were emphasized. The following metrics were considered while evaluating user experience for different modes: cognition, confidence, engagement, immersion, and satisfaction. A Likert scale ranging from 1 (poor) to 5 (excellent) was used to measure evaluation metrics. Participants were given 20 minutes in total to explore the terrain in different modes.

Ten sighted subjects (eight males and two females) aged 20 to 55 volunteered to participate in the user study. Users evaluated different evaluation metrics for the following modes in a VR-enhanced geographical topology exploration: 1) visual, 2) audio-visual (continuous), and c) audio-visual (sporadic). Here, the independent variable is the mode, and the dependent variable is the user evaluation score for different evaluation metrics. During the evaluation, users were asked to identify which mode(s) helped them understand the terrain's characteristics (concentrations of limestones, dolomites, and gypsum with different sensory parameters, i.e., color and pitch, based on their representations) most accurately. Users were further asked to identify the mode(s) that made them most confident with their diagnoses and provided the best immersive and engaging experience than others to motivate them to continue in-depth explorations. Satisfaction was related to the comfort and relaxation experienced with a particular mode.

Likert scale datasets are considered ordinal or ranked. As the normality of data was not confirmed for all options in the results, the non-parametric Friedman test was conducted to determine significant differences among the three options. Table 1 shows the Friedman test summary for different evaluation metrics. There were significant variations in subjective ratings with modes. Here are Friedman statistics (χ^2) and p-values for different evaluation metrics: cognizance ($\chi^2 = 8.67$, $p = 0.01 < 0.05$), satisfaction ($\chi^2 = 10.89$, $p = 0.00432 < 0.01$), confidence ($\chi^2 = 12.2$, $p = 0.00224 < 0.01$), engagement ($\chi^2 = 10.23$, $p = 0.006 < 0.01$), and immersion ($\chi^2 = 9.48$, $p = 0.00875 < 0.01$).

Table 1: Friedman test summary for different evaluation metrics

Metrics	Friedman statistic	P value	Kendall's coeff.	Effect size
Cognizance	8.67	0.01	0.43	Moderate
Satisfaction	10.89	0.00432	0.54	Large
Confidence	12.2	0.00224	0.61	Large
Engagement	10.23	0.006	0.51	Large
Immersion	9.48	0.00875	0.47	Large

The post hoc comparison was conducted using the Durbin-

Conover pairwise comparison test with Holm corrections. The user evaluation scores varied significantly with modes for different evaluation metrics. The bar charts in Figure 2 demonstrate variations in evaluation metrics with modes.

Audio-visual (sporadic) mode was preferred by most participants, followed by audio-visual (continuous) and visual modes. Here is a representative comment from a participant: "Audio-visual sporadic feels like it gives the most information about the model." Two participants commented about adding a way to increase the speed of changing spots in the sporadic mode. Overall, participants demonstrated a preference for multimodal visualization.

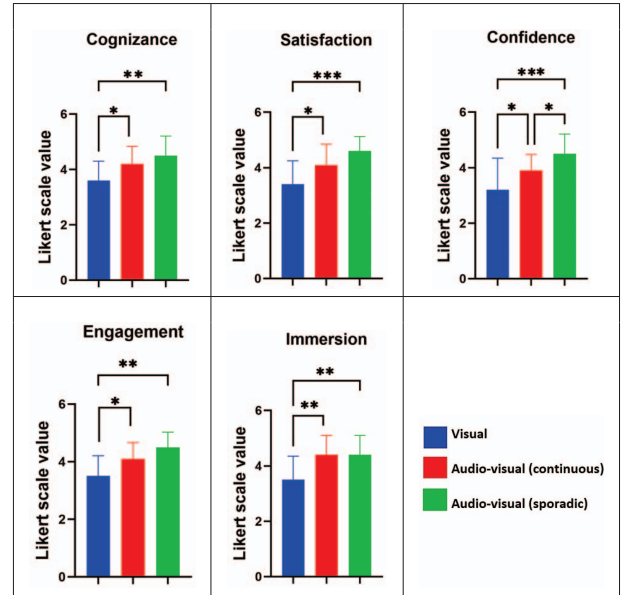


Figure 2: Bar charts comparing evaluation metrics for different modality options. The asterisks in the bar graphs show significant differences between modes. (* represents $p \leq 0.05$, ** represents $p \leq 0.01$, *** represents $p \leq 0.001$).

5 CONCLUSION AND FUTURE WORK

Future work will explore wider variations in geographical topology in an audio-visual VE. The sporadic mode will allow users to interactively select the change rate from one point to the next. The current implementation used one auditory parameter, i.e., pitch for the auditory display. Multiple auditory parameters (i.e., frequency, tempo, and more) will be used to enrich sound variations.

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